Shibboleth: Probe-and-Replay Model-Response Watermarking for Diffusion Language Models

Anonymous Author(s)
Affiliation
Address
email

Abstract

Most text watermarks encode a secret in the chosen tokens and later inspect the text alone. We study a different channel available in masked diffusion language models (DLMs): after generation, the same model can reveal how an emitted token's probability changes when selected parts of its context are hidden. This reproducible response ties the watermark to both the text and the checkpoint.

Shibboleth turns that response into a detector. It first finds positions whose candidate tokens span both response directions. Only after a position is selected does the key choose which direction to favour. The generator then resamples from the current model distribution, without changing the decoding schedule. Verification reconstructs the same masked contexts from the finished text and counts keyed sign agreements. Because position selection cannot observe the requested sign, text generated independently of the key produces an exact binomial count.

At 1% FPR, Shibboleth reaches 0.80 TPR on LLaDA-8B-Instruct and 0.90 on Dream-7B-Instruct. The strongest token-based baseline reaches 0.93 on LLaDA, but falls to 0.57 or below on Dream when its LLaDA setting is transferred unchanged. At Shibboleth's main setting, the PPL ratio is $0.74\times$ on LLaDA and $1.09\times$ on Dream, while measured task-score changes are at most 0.06. Verification requires the same checkpoint and 144 full-sequence model calls per 512-token document; dispersed edits reduce detection power.

1 Introduction

Language-model text can be difficult for readers to distinguish from human writing, creating provenance risks in disinformation, phishing, and academic-integrity review [28]. A post-hoc detector learns the output patterns of a generator; a watermark instead changes generation under a secret key and lets the key holder test a later document [21, 33]. We study watermarking because it permits a stated false-positive rate, including at finite sample size, rather than relying on a classifier whose behaviour may change with the decoder [10, 19, 23, 44].

Nearly all language-model watermarks place the secret in token identities: the key biases a vocabulary subset or couples randomness to each draw. Their detector therefore needs only the text and key (Figure 1a) [21, 24]. This design is inexpensive, but it must coexist with the decoder. For masked DLMs, that decoder may resolve many positions in parallel and choose their order from confidence, so forcing a resolved prefix or steering the order can change the system being watermarked [17].

Our starting observation is simple: a masked DLM can answer a reproducible question about its own output. Take a completed sentence, hide one subset of its earlier context, and record the probability of an emitted token; then hide a second subset and record it again. Some candidates become more likely and others less likely. The sign of that change is a *model response*. Because the verifier can recreate both hidings from the finished text, this response can carry a keyed signal even when the token identity alone does not (Figure 1b).

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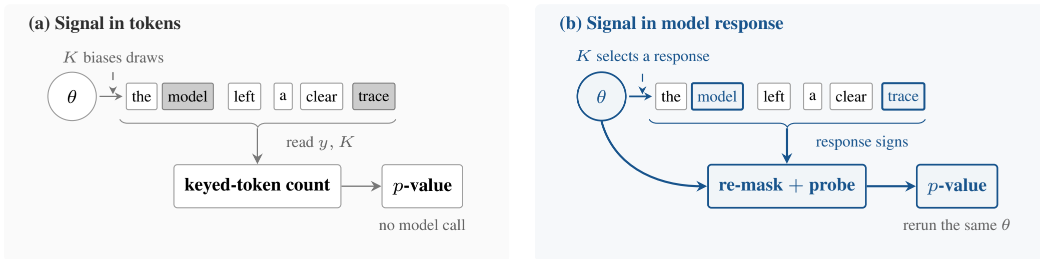

Figure 1: **Two places to encode a watermark.** Token-local methods alter sampled identities and verify from y, K alone. Shibboleth instead encodes which context response the checkpoint favours, so verification reruns the same θ.

Two details determine whether this idea is statistically sound. First, verification must recreate exactly the state used during generation, although later tokens are now visible. Second, a method may choose positions where both response directions are plausible, but it must do so before the key reveals which direction will count as a match. Otherwise position selection itself raises the false-positive rate.

Shibboleth addresses both points. For each active block, it re-masks that block and its suffix, compares pairs of hidden-prefix probes, and selects positions that retain probability mass on both response directions. The key then supplies the requested direction. At a selected position, Shibboleth draws up to R candidates from the decoder's current distribution and commits the first match; all other positions follow the reference decoder. Verification reconstructs the probes, repeats the selection rule, and evaluates an exact binomial tail. The block partition, confidence order, and checkpoint probabilities remain those of the reference decoder.

Our contributions are:

- We introduce model-response watermarking, where the signal is a checkpoint's reproducible reaction to hidden context rather than a keyed pattern in token identities.
- We give a replayable construction whose carrier positions are selected before the keyed orientation is consulted, yielding finite-sample false-positive control while preserving the reference decoding schedule (Sections 3.2–3.5).
- We evaluate two masked DLMs. At 1% FPR, Shibboleth reaches 0.80 TPR on LLaDA and 0.90 on Dream. Verification requires 144 full-sequence model calls, and detection power decreases under dispersed edits (Section 4).

2 Related Work

Masked diffusion language models. Masked DLMs generate text by repeatedly predicting unresolved positions and committing a subset of them. This extends discrete diffusion to text [3, 16, 29, 32, 37, 38] and now supports instruction-following models including LLaDA and Dream [25, 31, 42, 45]. Which positions commit, and in what order, affects both quality and speed [2, 6, 20, 26, 39, 40]. We therefore treat the deployed schedule as a constraint: Shibboleth observes the model's masked-token predictions but does not choose a new commit order.

Watermarks encoded in tokens. Autoregressive watermarks usually bias a keyed vocabulary subset [21] or couple each draw to keyed randomness [1, 13, 24]. Their detectors are cheap because a key–token calculation needs no model call, and later work characterises when those detectors are reliable or undetectable [10, 22, 44]. DLM methods adapt this idea by removing prefix dependence, averaging a red–green bias over unresolved context, sampling with keyed randomness, or encoding information in the denoising order [4, 9, 14, 17, 34, 41]. Our closest baseline, dgMARK, changes which mask resolves next and later reads that order from the text. Shibboleth instead keeps the order fixed and asks the checkpoint to reproduce its response.

Detectors that use a model. Model access can support richer tests than a key–token hash. Likelihood-ratio detectors compare watermarked and base conditionals [18]; access to the model can improve a Neyman–Pearson score over a key-only Gumbel statistic [12]; GaussMark verifies a keyed weight perturbation with a forward–backward pass [7]; and model-theft watermarks query a suspect system [43]. Outside watermarking, DetectGPT scores text using log probabilities from the model of interest and perturbed versions of the passage; replacing the source model with a surrogate weakens detection [30]. These model-assisted detectors are most natural when a provider or central verifier can afford model access. Shibboleth shares that operational setting but adds a keyed generation channel and a document-level test. It leaves the checkpoint unchanged and keys the contexts presented to it, so one deployed model can serve many keys.

Finite-sample detection and editing. Watermark tests increasingly replace asymptotic z-scores with exact tails, permutation tests, or corrected maxima over windows [12, 22, 24]. After editing, a statistic that concentrates on surviving regions can outperform a document-wide sum [27]. Shibboleth uses an exact binomial tail for the unedited detector and compares document-wide, block-local, and windowed aggregation under editing in Section 4.2.

3 Shibboleth: Watermarking by Model Response

3.1 Masked Decoding

A masked DLM does not have a unique "next token." It sees a sequence containing masks and returns a distribution at every unresolved position. Let x be the prompt, $y = (y_1, \dots, y_N)$ the continuation, and I the positions already revealed. At any masked position $i \notin I$, the model provides $p_\theta(y_i \mid y_I, x)$. One forward pass over the full sequence provides these distributions for every unresolved position; we count this as one full-sequence model call.

We study the blockwise schedule used by LLaDA [2, 31]. Blocks finish from left to right, while a confidence rule decides which masks inside the active block commit at each step. We write B_b for the active block and F_b for the completed prefix before it. Shibboleth keeps both the block order and this confidence rule.

3.2 Problem Setup

The goal is to distinguish text generated with a secret document key from text generated independently of that key, at a false-positive rate fixed in advance. The watermark may change which token is sampled, but it should not replace the model's decoder or its schedule.

What the two parties retain. The embedder has the model, prompt, secret key K, and target bits. The verifier later has the completed sequence, the same checkpoint, K, and public parameters—but no generation trace, stored logits, random seed, or record of rejected samples. A nonce derives a fresh document key $K_d = \mathrm{HMAC}(K, \mathrm{nonce}_d)$ [5]. Separate hash domains use K_d for probe construction, carrier assignment, and the requested response directions.

Guarantee and scope. For any text produced independently of K_d, including output from another sampler, a detector run at level α fires with probability at most α. This guarantee is finite-sample and permits the method to choose a data-dependent number of carriers. It does not promise detection after editing, key disclosure, adaptive detector queries [19], or replay with a numerically different checkpoint.

Why write during generation? A tempting alternative is to watermark only after the text is complete. Across 36 post-hoc substitution settings, however, the median selected position has only one candidate within a 3-nat likelihood budget, and the best setting below a $1.35\times$ PPL budget reaches 0.10 TPR at 1% FPR. Shibboleth therefore writes while the model's live distribution is available.

The design follows three requirements:

1. **Replayability.** From the completed sequence, the verifier must reconstruct the same masked context that the embedder used.
2. **Blind selection.** Carrier positions must be fixed before the key reveals which response direction counts as a match.

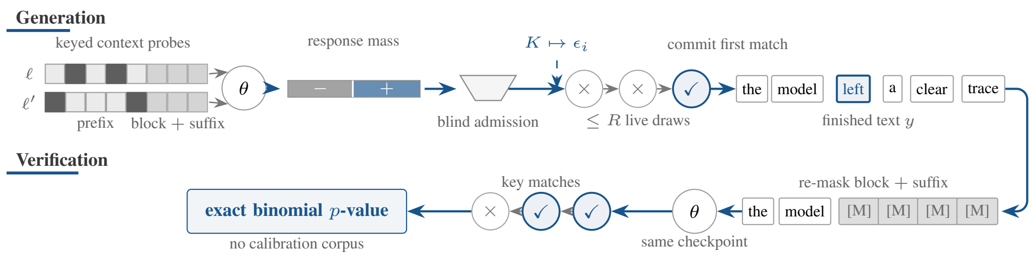

Figure 2: **One block of generation and verification.** Two hidden-prefix views separate candidates by response direction. Carrier selection occurs before the key supplies its orientation; generation commits the first matching draw. Verification rebuilds the same masked state and returns an exact binomial p-value.

3. **Schedule preservation.** Blocks and within-block commit order must remain those of the reference decoder.

Replayability makes a generation-time response measurable later; blind selection keeps data-dependent carrier discovery from changing the stated false-positive rate. The next three subsections implement these requirements.

3.3 Probing the Model and Selecting Carriers

For each unfinished block, Shibboleth asks the model the same question under several hidden-prefix views: how likely is each candidate token when this part of the earlier context is missing? A candidate's response is the change between two such views. A useful carrier is a position where the model still assigns probability mass to candidates with both response directions (Figures 2 and 3).

Formally, Shibboleth constructs L pseudorandom probe patterns $D_{b,\ell} \subset F_b$, where each pattern specifies prefix positions to hide. It also masks the current block B_b and every later position. One model evaluation for pattern ℓ then gives the log-probability of vocabulary item v at position i:

$$\lambda_{i,\ell}(v) = \log p_\theta(v \mid F_b \setminus D_{b,\ell}; B_b \text{ and suffix masked})\,. \tag{1}$$

The block and suffix are hidden because neither was available when the probe was created. The verifier can therefore reconstruct the same state from the completed sequence instead of seeing later tokens that would alter the probabilities. Probing a block costs $L+1$ full-sequence model calls: L hidden-prefix views and one view with no prefix positions hidden.

For a pair (ℓ, ℓ'), define the response $g_i^{\ell,\ell'}(z) = \lambda_{i,\ell'}(z) - \lambda_{i,\ell}(z)$. Let p_i^{base} be the block-masked distribution with no hidden prefix. The probability mass on either response direction, and a symmetric score for the pair, are

$$q_{i,\pm}^{\ell,\ell'} = \sum_z p_i^{\text{base}}(z)\mathbf{1}[\pm g_i^{\ell,\ell'}(z) > 0], \qquad S_i^{\ell,\ell'} = 2\min(q_{i,+}^{\ell,\ell'}, q_{i,-}^{\ell,\ell'}). \tag{2}$$

The score is high only when both directions remain available. At each position, Shibboleth keeps the pair with the largest S_i and selects the position when $S_i > s_{\min}$, subject to an optional per-block cap. Choosing the best pair raises the measured mean score from 0.28 for a fixed near/far pair to 0.45.

Crucially, swapping ℓ and ℓ' leaves S_i unchanged while reversing every response sign. Carrier selection therefore reveals nothing about which sign will later be rewarded. Figure 3 makes this ordering explicit.

After selecting the carrier, a separate hash domain supplies $\epsilon_i \in \{-1, +1\}$. A target $w_i \in \{-1, +1\}$ represents the provenance or payload bit. Candidate z counts as a match when

$$m_i(z) = \mathbf{1}[w_i \epsilon_i g_i(z) > 0]. \tag{3}$$

If $g_i(z) = 0$, a second domain-separated bit assigns the outcome. Such a tie cannot be steered, but it remains an independent fair outcome for detection.

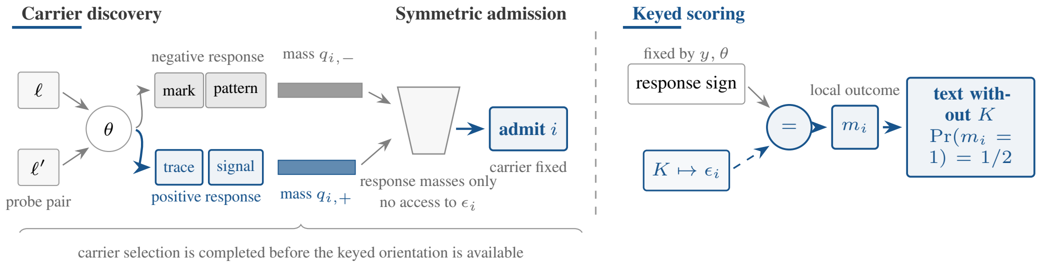

Figure 3: **Selection precedes the keyed orientation.** The probe pair divides candidates by response direction. Position i is selected from the two masses alone; only then does K provide ϵ_i. For text produced independently of K, the two signs agree with probability $1/2$. Token labels and bar widths are schematic.

3.4 Writing the Requested Answer

The reference decoder still chooses the next position and supplies its live distribution p_i. Unselected positions are sampled once as usual. At a selected carrier, Shibboleth accepts a candidate only if it has the requested response and is not too far below the most likely token:

$$A_i = \left\{ z : w_i \epsilon_i g_i(z) > 0 \ \text{ and } \ p_i(z) \geq \kappa \max_v p_i(v) \right\}. \tag{4}$$

It draws from the same p_i up to R times and commits the first member of A_i. If every proposal misses, it takes one fresh unchecked draw. All retries reuse p_i, so they add no model call and do not alter the commit order.

Let $a_i = p_i(A_i)$ be the mass that passes both conditions, and let $q_i = \sum_z p_i(z) m_i(z)$ be the chance that an unchecked draw scores as a match. The committed token comes from $p_i(\cdot \mid A_i)$ with probability $1 - (1 - a_i)^R$ and from p_i otherwise. Its match probability is therefore

$$\Pr[m_i = 1] = 1 - (1 - a_i)^R (1 - q_i). \tag{5}$$

With no plausibility floor and no ties, $a_i = q_i$, giving $1 - (1 - q_i)^{R+1}$. Increasing R strengthens detection but changes the sampling distribution; κ limits how unlikely a guided proposal may be relative to the live argmax. It does not constrain the unchecked fallback or guarantee semantic quality, which we measure separately.

3.5 Watermark Detection

The verifier re-masks each completed block and suffix, reconstructs the probe pairs and carriers, and evaluates $m_i(y_i)$. It needs no rejected samples or generation trace. For provenance detection, the selected carriers form a pool P with $w_i = +1$; a payload implementation would keep a separate pool for each message bit. The document statistic and its one-sided p-value are

$$T = \sum_{i \in P} m_i, \qquad p_{\text{det}} = \Pr\left[\text{Binomial}(|P|, \tfrac{1}{2}) \geq T\right]. \tag{6}$$

Figure 4 follows a concrete five-carrier example from the replayed outcomes to the exact upper-tail probability.

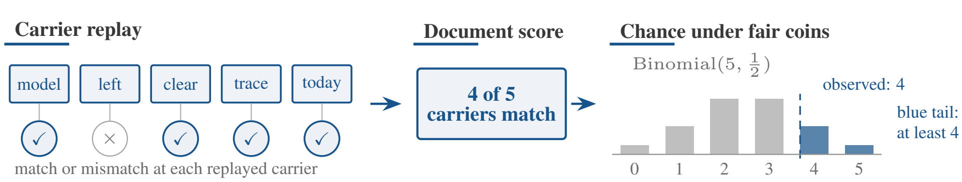

Figure 4: **Document-level detection.** The verifier replays each carrier and records a match. Here four of five match; the blue bars give the exact chance of observing at least four matches under fair coins.

Proposition 1 (Exact false-positive control)**.** Assume the hash domains behave as independent random functions. For any text written without the document key, condition on the selected carriers,

Method	Setting	PPL ratio ↓	TPR at FPR ↑ 1%	0.1%	Full-sequence model calls ↓ Generation	Verification
LLaDA-8B-Instruct						
KGW [21]	$\delta = 1$	1.22×	0.90	0.83	512	**0**
dgMARK [17]	$k = 1$	1.45×	0.83	0.80	512	**0**
dgMARK [17]	+3-beam	1.21×	**0.93**	**0.93**	∼2048	**0**
Shibboleth	$R = 1$	0.97×	0.43	0.20	656	144
Shibboleth	$R = 8, \kappa = 0.1$	0.87×	0.70	0.60	656	144
Shibboleth	$R = 16, \kappa = 0.05$	**0.74**×	0.80	0.60	656	144
Dream-7B-Instruct						
KGW [21]	$\delta = 1$	1.84×	0.57	0.37	512	**0**
dgMARK [17]	$k = 1$	1.75×	0.37	0.27	512	**0**
dgMARK [17]	+3-beam	**0.71**×	0.30	0.30	∼2048	**0**
Shibboleth	$R = 1$	1.34×	0.40	0.27	656	144
Shibboleth	$R = 8, \kappa = 0.1$	0.79×	0.70	0.67	656	144
Shibboleth	$R = 16, \kappa = 0.05$	1.09×	**0.90**	**0.90**	656	144

Table 1: **Detectability and cost at 512 tokens.** PPL ratio is measured against each method's own unwatermarked run and should be interpreted within that method. Shibboleth uses $n = 30$ documents ($n = 20$ for $R = 16$); Dream PPL uses 8–13 successfully scored document pairs per setting. Settings chosen on LLaDA are transferred to Dream without retuning. One model call is one forward pass over the current full sequence; these counts measure algorithmic work, not runtime. δ is KGW's logit bias and k is dgMARK's candidate count.

probe pairs, responses, and target bits. The orientation bits remain independent fair coins. Hence the m_i are independent $\mathrm{Bernoulli}(\frac{1}{2})$ variables, $T \sim \mathrm{Binomial}(|P|, \frac{1}{2})$, and thresholding Eq. (6) at α gives false-positive rate at most α for every document length and carrier count.

Why adaptive carrier selection is allowed. The text and model may determine how many carriers exist, which probe pair is chosen, and how large each response is. All of those choices are fixed before the independent orientation bit is used. Flipping ϵ_i then flips the match whenever $g_i(y_i) \neq 0$; the tie rule supplies a fresh fair bit otherwise. The detector therefore needs neither an asymptotic approximation nor a calibration corpus. The guarantee still assumes text produced independently of the key and domain separation among the key's several roles.

4 Experiments

Models and data. We evaluate LLaDA-8B-Instruct [31] and Dream-7B-Instruct [42] in FP16 on one TITAN RTX. Detectability uses C4 `realnewslike` prompts [36], truncated to 300 characters, and 512-token continuations divided into 32-token blocks. We correct Dream's one-position logit offset in its wrapper, then use the same position convention for generation and replay.

Decoders and settings. Shibboleth and dgMARK use each model's confidence schedule with multinomial sampling at temperature 0.8; dgMARK additionally uses top-$k = 3$. KGW requires strictly left-to-right decoding. Shibboleth uses $L = 8$ probe patterns, threshold $s_{\min} = 0.5$, retry budget R, and optional plausibility floor κ. We evaluate provenance ($w_i = +1$), not payload recovery. Here one model call means one forward pass over the current full sequence. With 16 blocks and $L + 1 = 9$ probe views, verification replays $16 \times 9 = 144$ calls; generation uses 512 ordinary decoding calls plus the same 144 probe calls, for 656 in total.

Metrics. Detection is TPR at the analytic FPR thresholds in Eq. (6). Perplexity is scored by GPT-2-large [35] as $\exp(\mathrm{mean}(\mathrm{NLL}_{\mathrm{wm}} - \mathrm{NLL}_{\mathrm{unwm}}))$. A PPL ratio therefore compares a method with its own unwatermarked decoder; it is not a semantic-quality score and is only indicative across different decoders. The baselines are KGW [21] and dgMARK [17].

4.1 Main Results and Analysis

Retries trade sampling distortion for detection. Table 1 gives the 512-token detection, quality, and model-call comparison. On LLaDA, $R = 1$ reaches 0.43 TPR at 1% FPR. On 30 held-out

Method	MMLU Acc ↑	GSM8K Acc ↑	HumanEval Pass@1 ↑
LLaDA-8B-Instruct			
KGW [21]	0.530→0.535	0.640→0.740	0.329→0.293
dgMARK [17]	0.605→0.530	0.740→0.660	0.409→0.341
dgMARK +3-beam	0.605→0.530	0.740→0.660	—
Shibboleth	0.595→0.575	0.640→0.700	0.335→0.384
Dream-7B-Instruct			
KGW	0.640→0.625	0.700→0.700	—
dgMARK	0.740→0.570	0.840→0.560	—
dgMARK +3-beam	0.740→0.645	0.840→0.560	—
Shibboleth	0.695→0.710	0.660→0.680	—

Table 2: **Task accuracy before and after watermarking.** Each cell is unwatermarked → watermarked under that method's decoder. MMLU uses 200 questions, GSM8K 50 problems, and HumanEval 164 executed tasks. Approximate standard errors are 0.035, 0.07, and 0.037, respectively; — denotes an unmeasured pair. Shibboleth uses $R = 16, \kappa = 0.05$.

prompts, $R = 2, 4, 8$ increase TPR to $0.73, 0.83, 0.87$, consistent with Eq. (5). Retries add no model evaluation, but without a floor the PPL ratio rises from $1.14\times$ at $R = 2$ to $1.61\times$ at $R = 8$. The floor removes low-probability guided draws: $(R, \kappa) = (8, 0.1)$ gives 0.70 TPR at $0.87\times$ PPL, and $(16, 0.05)$ gives 0.80 at $0.74\times$. Ratios below one mean only that GPT-2-large prefers the retained high-probability tokens.

The same settings behave differently across checkpoints. We transfer all LLaDA settings to Dream without retuning. KGW falls to 0.57 TPR at 1% FPR and produces 84% duplicate bigrams under forced left-to-right decoding. Direct and three-beam dgMARK reach 0.37 and 0.30 TPR. Shibboleth at $(16, 0.05)$ reaches 0.90 at both reported FPR levels. Its PPL ratio is $0.79\times$ at $(8, 0.1)$ and $1.09\times$ at $(16, 0.05)$, based on 8–13 valid pairs per arm. Detection transfers here; the PPL-minimising setting does not.

Task accuracy exposes decoder-specific costs. Table 2 shows each method's own unwatermarked and watermarked scores side by side; this pairing matters because each method uses a different decoding procedure. For Shibboleth, changes are at most 0.06 across the measured MMLU, GSM8K, and HumanEval cells. For direct dgMARK, HumanEval on LLaDA changes from 0.409 to 0.341; on Dream, MMLU changes from 0.740 to 0.570 and GSM8K from 0.840 to 0.560. These task outputs are also shorter: on LLaDA GSM8K, Shibboleth and direct dgMARK reach 0.12 TPR at 1% FPR, three-beam dgMARK 0.44, and KGW zero. The 512-token detection rates therefore do not describe 256-token answers.

Model-response detection costs model calls. At 512 tokens, three-beam dgMARK reaches 0.93 TPR at both 1% and 0.1% FPR, direct dgMARK $0.83/0.80$, KGW $0.90/0.83$, and Shibboleth $0.80/0.60$. The token-local methods need no model evaluation during detection; Shibboleth needs 144. KGW's left-to-right decoder also repeats heavily: before bigram deduplication, 37% of its unwatermarked documents cross the decision threshold; after deduplication, two of thirty still cross the 0.1% threshold. With cohorts of 20–30 documents, TPR differences near 0.1 remain imprecise.

4.2 Calibration and Robustness

Observed false positives remain below the requested levels. Across 20 keys, with 120 unwatermarked LLaDA generations per key, mean FPR is 0.0121, 0.0037, and 0.0000 at nominal 0.10, 0.05, and 0.01. The largest per-key rates are 0.0583, 0.0250, and 0.0000. Every Dream unwatermarked cohort has zero false positives at 1%.

Editing mainly breaks prefix synchronisation. At $R = 1$, either same-model argmax re-denoising or deleting a random 10% of positions reduces TPR at 1% FPR from 0.35 to 0.05 on 20 documents. At $R = 16, \kappa = 0.05$ and 1024 tokens ($n = 16$), a random 5% edit leaves TPR at 0.88, while a random 10% edit reduces it to 0.50; false-positive control remains valid on identically edited unwatermarked text. In contrast, re-denoising one contiguous span covering 10% or 30% of positions leaves TPR at 0.88, because carriers before the span are unchanged.

Neither tested aggregation change improves the random-edit result. A Bonferroni-combined per-block test gives 0.38 TPR versus 0.50 for document-wide counting under random 10% editing, and 0.69 versus 0.88 without editing. A 128-token conditioning window gives 0.44 versus 0.50 under random 10% editing. Copy–paste-style sparse survival was not evaluated.

5 Discussion

Deployment role. Shibboleth is designed for model providers, centralised verification services, and auditors of high-value documents: settings in which the production decoder should remain unchanged and a verifier can retain the checkpoint. It is not a replacement for inexpensive token-local detection. A practical system could use a token-local test for routine traffic and reserve checkpoint replay for cases where model-specific provenance matters. This binding is both the method's distinction and its cost. The checkpoint identity, numerical precision, and replay implementation must all be retained; our experiments do not establish transfer across numerically different copies.

Interpreting a detection result. A low p-value supports a narrow statement: under the registered key and replay procedure, the document contains more checkpoint-consistent carrier responses than expected by chance. It does not identify the person who requested the text, establish that every passage came from the model, or rule out later editing. Those questions require account records, cryptographic signatures, or separate forensic analysis. A verifier should report the p-value and carrier count, along with the key identifier, checkpoint hash, numerical precision, probe construction, and decision threshold. The carrier count helps distinguish an inconclusive short document from a response that disagrees with the checkpoint.

Editing and quality. Dispersed edits alter many later prefixes and therefore many reconstructed probes; a contiguous span leaves all earlier carriers intact. This asymmetry points to content-anchored probes, rather than absolute prefix positions, as a promising route to editing tolerance. The block-local and windowed variants tested here did not improve detection after random edits. Quality measurements also require restraint: a PPL ratio below one does not imply better text, and the task cohorts are too small to support claims about modest score changes. Under the measured settings, Shibboleth's task-score changes are small relative to the decrease observed for direct dgMARK on Dream, but its verifier requires substantially more computation.

6 Conclusion

Shibboleth shows that a masked DLM can participate in authenticating its own output. The watermark is carried by reproducible changes in model probability under hidden context, rather than by a keyed token subset. Its central design choice is temporal: select a carrier from a symmetric response score first, then reveal the keyed orientation. This ordering gives an exact binomial test even though the text and model determine how many carriers are available.

The result complements token-local watermarking rather than replacing it. Shibboleth preserves the deployed decoding schedule, and its TPR rises from 0.80 on LLaDA to 0.90 on Dream under settings transferred without retuning. Its verifier, however, must rerun the same checkpoint 144 times per 512-token document. Token-local detectors remain cheaper and, on LLaDA, reach higher TPR.

Limitations. The false-positive guarantee does not imply detection after editing. Random 10% re-denoising reduces TPR from 0.35 to 0.05 at $R = 1$ and from 0.88 to 0.50 at $R = 16$, although one contiguous edited span leaves the measured TPR unchanged. Replay also requires the same checkpoint and numerical precision. Evaluation covers two masked DLMs, one prompt corpus, cohorts of 16–50 documents, and provenance only; payload recovery is not measured.

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A Additional Tables

Table 3 reports the measured per-key false-positive rate of the deployed decision rule. Table 4 summarises the post-editing evaluation under the deployed pooled detector. Table 5 expands Table 2 into the full task-accuracy grid, including the greedy-decoding cells.

Nominal α	Mean FPR	Highest per-key FPR
0.10	0.0121	0.0583
0.05	0.0037	0.0250
0.01	0.0000	0.0000

Table 3: **Observed false-positive rates across keys.** The evaluation uses 20 keys and 120 unwatermarked generations per key under the deployed decision rule.

		TPR at FPR ↑	
Attack	Strength	1%	0.1%
Configuration A: $R = 1$, 512 tokens ($n = 20$)			
None	—	0.35	—
Re-denoising (same model, argmax)	10% of positions	0.05	—
Deletion	10% of positions	0.05	—
Insertion	10% of positions	—	—
Substitution (random)	10% of positions	—	—
Copy–paste	25% kept, 1 segment	—	—
Paraphrase (DIPPER) [23]	lex 20, order 0	—	—
Configuration B: $R = 16, \kappa = 0.05$, 1024 tokens ($n = 16$)			
None	—	0.88	0.88
Re-denoising (random positions)	5% of positions	0.88	0.75
	10% of positions	0.50	0.38
Re-denoising (contiguous span)	10% of positions	0.88	0.88
	30% of positions	0.88	0.88

Table 4: **Detection after editing.** The pooled detector uses the same analytic thresholds as Table 1. Random edits disrupt later prefixes; a contiguous span preserves all earlier carriers. Because the cohorts are small, one document changes TPR by 0.05 when $n = 20$ and about 0.06 when $n = 16$. An em dash denotes an unmeasured setting.

		Greedy decoding			Multinomial decoding		
Model	Method	MMLU Acc ↑	GSM8K Acc ↑	HumanEval Pass@1 ↑	MMLU Acc ↑	GSM8K Acc ↑	HumanEval Pass@1 ↑
LLaDA-8B-Instruct	Non-watermarked	0.605	0.740	0.409	0.595	0.640	0.335
	KGW [21]	—	—	—	0.535	0.740	0.293
	dgMARK [17]	—	—	—	0.530	0.660	0.341
	dgMARK +3-beam	—	—	—	0.530	0.660	—
	Shibboleth	—	—	—	0.575	0.700	0.384
Dream-7B-Instruct	Non-watermarked	0.740	0.840	—	0.695	0.660	—
	KGW	—	—	—	0.625	0.700	—
	dgMARK	—	—	—	0.570	0.560	—
	dgMARK +3-beam	—	—	—	0.645	0.560	—
	Shibboleth	—	—	—	0.710	0.680	—

Table 5: **Text generation quality, full grid.** Accuracy on MMLU [15] and GSM8K [11] and pass@1 on HumanEval [8] under greedy and multinomial decoding, with the unwatermarked run of each model as the reference row of its block. Shibboleth uses $R = 16, \kappa = 0.05$; 50 problems per GSM8K cell, so differences in individual scores should be interpreted cautiously. Each watermark decodes under its own regime—KGW left-to-right (its green list requires a resolved prefix; its own control scores 0.640), dgMARK under confidence decoding (its argmax control is the greedy reference cell)—so rows are read against their own controls, not across. An em dash denotes an unmeasured setting.